

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2020

Bachelor in Computer Applications

Course Title: **Computer Fundamentals & Applications**

Code No: **CACS101** Semester: **I**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Who is a database administrator? Explain the major roles and responsibilities of DBA.
3. Define operating system. Explain the open-source operating system.
4. What is an ergonomic keyboard? Explain the working mechanism of a keyboard.
5. Explain different types of computers on the basis of size and power.
6. What is DBMS? Explain the Relational database model with a suitable diagram.
7. Prepare a specification of a personal computer which you want to purchase.
8. Explain any five DOS commands with suitable examples..

Group C

Attempt any TWO questions[2x10=20]

9. Prepare the following spreadsheet:

Student Name	CF	Logic	Math	So	Tech	Total	Per	Result	Grade	Rank	Remarks
1Ram	45	56	54	44	45	220	44	???	???	???	???
2Sham	50	45	53	45	45	220	44	???	???	???	???
3Hari	46	44	44	44	44	182	36	???	???	???	???

Conditions: a) Write an appropriate formula to find the total marks of each student. b) Write a formula to find out the percentage of each student. c) Result is "Passed" if the student scores at least 40% marks in each subject. d) Write a formula to calculate the grade of the student based on percentage: - A for $\geq 90\%$ - B for $\geq 80\%$ - C for $\geq 70\%$ - D for $\geq 60\%$ - E for $\geq 40\%$ e) Write a formula to find the rank of individual students who have passed. f) Write remarks based on the following conditions: - "Excellent" for Grade A - "Good" for Grade B - "Fair" for Grade C - "Moderate" for Grade D - "Satisfactory" for Grade E
10. a) What is mail merge? Explain the major functions available in the Review tab of a Word document. b) What is Photoshop? Write the steps to create an MRP-sized photograph in JPEG format.
11. Define Computer Network. Explain the OSI model in detail.